

Total No. of Questions : 8]

SEAT No. :

PB2505

[Total No. of Pages : 2

[6263]-391

B.E.(AI&DS)

DISTRIBUTED COMPUTING

(2019 Pattern)(Semester - VIII)(417531)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Neat diagrams must be drawn Wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use electronic pocket calculator is allowed.*

- Q1)** a) List and explain any one variant of Paxos in detail. [6]
b) Explain Fault Tolerance and Recovery in context of Distributed systems. [6]
c) Explain reinforcement Learning for Dynamic Load Balancing. [6]

OR

- Q2)** a) What is consensus algorithms? Explain any one algorithm. [6]
b) Explain Genetic Algorithms for Task Scheduling. [6]
c) Compare Centralized Load Balancing & Distributed Load Balancing Techniques. [6]

- Q3)** a) Explain Systems and Architectures for Distributed Machine Learning. [9]
b) Write note on [8]
i) Federated Learning,
ii) Hogwild
iii) Elastic Averaging SGD

OR

- Q4)** a) What is Apache Spark? Explain working of Apache Spark. [9]
b) Explain how integration of AI algorithms in distributed systems can help in Intelligent Resource Management, Anomaly Detection. [8]

P.T.O.

- Q5)** a) Explain the Big data processing frameworks in distributed computing. [6]
b) Differentiate between SIMD and MIMD. [6]
c) Elaborate various scalable data ingestion methods used in distributed computing environments. [6]

OR

- Q6)** a) Explain how AI and data science can be applied for large-scale data processing and analytics. [6]
b) Compare SIMD and MISD [6]
c) Discuss about various types of real-time analytics used in distributed computing systems. [6]

- Q7)** a) Explain Anomaly as well as Behavior AI-based Intrusion Detection&Threat Mitigation Techniques [9]
b) Explain how can Secure Multi-Party Computation(SMPC)be effectively implemented to ensure confidentiality and privacy preservation. [8]

OR

- Q8)** a) Enlist the various security challenges in distributed systems? Elaborate any three challenges in detail? [9]
b) Write a Short Note a Threat Hunting and Visualization. [8]
